Sourav Singh

Software Engineer, Backend and Distributed Systems

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SUMMARY

Java backend engineer with 4+ years building event-driven microservices and data systems, at home with both high-volume batch data and fast-moving streams: ~70 PB on LinkedIn's data platform and ~40K writes/sec through Kafka pipelines at Cashfree.

EXPERIENCE

Software Engineer
April 2024 – April 2026

LinkedIn · Unified Metrics Platform (Data Infrastructure)

- Led the engineering response to an org-wide Code Yellow, building a release process with automated regression, a 24-hour soak and gated promotion. Zero hard-SLA misses over 3 months.
- Decoupled LinkedIn's A/B testing reporting into a standalone library used by another team, owning the interface design and the handover; cut platform storage 7% and memory usage 4%.
- Built an LLM tool that checks new pipelines' output and flags regressions, cutting onboarding from 3 days to 1.
- Built a duplicate-pipeline detector (~98% accurate) using embeddings and vector search, so teams reused existing pipelines instead of rebuilding them.
- Set up LinkedIn infra's first central testing platform, where upstream teams tested changes against real pipelines before release; adopted by 6+ teams. Recorded an internal series on debugging methods.
- Helped build LinkedIn's company-wide data platform on dbt and Iceberg, ~70 PB across 7,000+ daily flows, moving 4,000+ live pipelines onto it while cutting Spark usage 15%.
- Built APIs that let pipelines read Hive and Iceberg interchangeably, enabling migration of 15,000+ tables without an SLA breach.

Software Development Engineer (SDE-1)
January 2022 – April 2024

Cashfree Payments

- Architected an event-driven pipeline (MySQL to Kafka to Elasticsearch, with ActiveMQ retries and a dead-letter queue) sustaining ~40K writes/sec for merchants including IRCTC and Swiggy.
- Built queue-backed asynchronous job processing with a worker pool and futures, cutting report generation from 5-6 hours to seconds.
- Kept payments fast through IPL peaks by warming the JVM on startup, so the hot execution path was JIT-compiled before a pod accepted traffic.
- Built a real-time merchant dashboard and REST APIs serving 3K QPS at p99 200ms, plus no-code faceted filtering over ~3,000 Elasticsearch fields.
- Kept bounce rates under AWS SES throttling thresholds with a MongoDB-backed check that suppressed repeat sends to over-messaged users.
- Engineered a backup reporting path so merchants kept access to their data during Elasticsearch outages.
- Right-sized Kubernetes resources from observed utilization to cut cost.

PROJECTS

Kapitsa · a fine-tuned LLM that writes original 'funda'-style quiz questions, aiming to be India's first central, searchable home for them (personal project)

- Built the training pipeline: an LLM judge separates reasoning puzzles from recall trivia to train a fine-tuned Qwen3-8B model, with Qdrant for search.

SKILLS

Languages & Backend: Java, Python, SQL, Bash, Scala · Spring Boot, microservices, REST APIs, concurrency, JVM memory and JIT behaviour

Data & Infra: Kafka, ActiveMQ, Spark, Airflow, dbt, Iceberg, Trino · MySQL, MongoDB, Elasticsearch, Redis · AWS (S3, SES), Kubernetes, Docker, CI/CD

AI tooling: Claude Code, Cursor, agentic workflows, RAG and embeddings, LLM validation tooling

EDUCATION

B.Tech, Electronics and Communications Engineering
2018 – 2022

National Institute of Technology, Jamshedpur